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(57) Abstract :

The present invention relates to a system and method for star pattern recognition for spacecraft navigation using neural networks. The system is configured to determine spacecraft position and attitude by processing star field images captured by an onboard imaging device. The captured image is provided as input to a trained neural network adapted to extract characteristic star features and identify corresponding star patterns by matching them with a stored star catalog. The proposed method reduces computational complexity and resource consumption compared to conventional star identification techniques, thereby enabling efficient operation on resource-constrained platforms such as CubeSats and small satellites. The invention facilitates accurate, autonomous, and reliable spacecraft navigation with improved processing speed and reduced error, making it applicable to a wide range of space missions..

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